Article

Optimal Covering Changepoint Analysis

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Abstract

Detecting changepoints in time series is a fundamental task in statistical modeling and data-driven decision-making. We introduce a novel set covering-based model for changepoint detection that leverages combinatorial optimization to identify an optimal set of segments explaining the observed data. Unlike conventional methods based on dynamic programming (DP), which often impose strict structural constraints on the objective function (e.g., additivity), our formulation uses Mixed-Integer Linear Programming (MILP). This approach offers significantly increased expressiveness and flexibility in the cost structure, accommodating diverse, non-additive loss functions and complex penalty schemes. Our formulation guarantees global optimality under this flexible cost structure, a critical advantage over heuristic or approximate approaches. The model's design enables high adaptability to different application domains, including finance, bioinformatics, and industrial monitoring. The increased efficiency of standard MILP solvers, coupled with tailored dominance rules, ensures practical scalability, achieving computational efficiency on datasets with several thousands of observations. Extensive experiments demonstrate that our approach outperforms state-of-the-art methods in both accuracy and runtime, offering a robust and interpretable solution for optimal changepoint detection.

Keywords: Time series, Set covering, segmentation, change point detection.

1. Introduction

Changepoint detection (CPD) is the task of identifying abrupt changes in the statistical properties of a time series and constitutes a foundational problem across statistics, machine learning, and signal processing [1–3]. The objective of CPD is to partition a temporal sequence into contiguous, non-overlapping segments, each characterized by internally homogeneous behavior. Such segmentation enables a clearer understanding of the underlying data-generating process and facilitates the application of targeted analytical or predictive models at the segment level. As a result, effective changepoint detection plays a critical role not only in descriptive analysis, but also in forecasting, anomaly detection, and data-driven decision-making.

CPD plays a critical role in a wide range of application domains, including financial market analysis (regime shifts), industrial process monitoring (fault detection), climate science (trend changes), and bioinformatics (genomic variation analysis). Across these domains, a CPD method must satisfy two key requirements: (i) accurately segment the data so as to faithfully represent the underlying process, and (ii) provide a globally optimal solution, since suboptimal segmentation can propagate errors downstream and lead to significant real-world costs.

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Time series segmentation has therefore attracted extensive attention, leading to the development of a broad spectrum of methodologies, including classical statistical techniques, Bayesian approaches, machine learning algorithms, and hybrid models. Among these, segmentation based on linear models remains particularly prominent due to its computational efficiency and interpretability. This setting, commonly referred to as piecewise linear regression [4], assumes that the time series can be approximated by a sequence of linear trends, each corresponding to a distinct segment.

For exact and globally optimal changepoint detection under additive cost structures, the dominant methodological paradigm is dynamic programming (DP). Methods such as Optimal Partitioning exploit the recursive structure of the problem, constructing the optimal segmentation for a sequence of length N from optimal solutions to shorter prefixes. This approach yields polynomial-time algorithms under suitable assumptions and has proven highly effective in practice. However, the DP framework relies fundamentally on the additivity of segment-wise costs, typically combined with a penalty proportional to the number of segments.

This additive and recursive structure imposes a significant modeling limitation. In many practical settings, the analyst wishes to enforce global structural constraints that couple properties of all selected segments. Examples include limiting the total duration of short segments, bounding the number of changepoints within specific time windows, or enforcing regularity in segment lengths to avoid excessively fragmented solutions. While certain such constraints can, in principle, be approximated through state augmentation or penalty tuning, doing so rapidly compromises the tractability and scalability of DP-based methods. As a result, DP frameworks remain largely confined to locally additive objectives, restricting their applicability in scenarios where global control over the segmentation structure is essential.

To overcome this limitation, we propose a novel formulation of the CPD problem as a combinatorial optimization model, specifically a set covering problem. In this formulation, all feasible segments are pre-enumerated and treated as decision variables, and the task of changepoint detection becomes the selection of a minimum-cost subset of segments that exactly covers the time series. Unlike breakpoint-indicator or network-flow formulations, this approach directly models segments as atomic objects, naturally enabling rich interactions among them.

The key advantage of the MILP framework lies not only in its ability to reproduce the classical additive cost structure, but more importantly in its capacity to enforce complex global constraints through simple linear inequalities. These constraints can link arbitrary subsets of segment variables, control aggregate properties such as the number, length, or distribution of segments, and encode domain-specific structural requirements that are difficult or impractical to express within DP-based models. Unlike breakpoint-indicator or network-flow formulations, the proposed set-covering approach provides a direct and flexible mechanism for global control over the segmentation. While MILP solving is NP-hard in general, recent advances in solver technology, together with problem-specific preprocessing and bounding techniques, make this approach computationally viable for problem sizes of practical interest.

In this paper, we introduce Set-Covering based Changepoint Detection (SC-CPD), a novel MILP framework for optimal time series segmentation. Our main contributions are as follows:

A set-covering MILP formulation for CPD with global structural control. We formulate the changepoint detection problem as a set covering model, enabling the direct incorporation of global constraints that are beyond the practical reach of dynamic programming methods.

Enhanced modeling fidelity through global constraints. We demonstrate the practical value of the proposed framework by introducing and analyzing several global constraints that improve the interpretability, robustness, and domain relevance of the resulting segmentations.

Scalability through preprocessing and dominance rules. We develop specialized preprocessing techniques, including dominance rules and bounding strategies, that significantly reduce the number of candidate segments and allow the model to scale to time series with several thousand observations.

Competitive computational performance and solution quality. Through extensive computational experiments, we show that SC-CPD produces higher-quality segmentations while achieving runtimes that are comparable to, and in many cases competitive with, state-of-the-art DP-based and heuristic methods.

The paper further illustrates the versatility of the proposed approach through a range of real-world applications, including financial time series, epidemiological data, and environmental monitoring. Common challenges such as noise, irregular sampling, and high-dimensional feature spaces are discussed, highlighting the advantages of MIP-based changepoint detection in addressing domain-specific structural requirements.

The remainder of this paper is organized as follows...

2. Related literature

The problem of optimal changepoint detection and time series segmentation has been extensively studied, and a wide range of exact and approximate methods have been proposed. Comprehensive surveys are provided in [2,3], which classify approaches into statistical, Bayesian, algorithmic, and learning-based families. In this section, we focus specifically on exact optimization-based formulations for changepoint detection and highlight their modeling assumptions and limitations relative to the proposed approach.

Dynamic programming (DP) constitutes the dominant paradigm for exact changepoint detection under additive cost structures. Classical methods such as Segment Neighborhood [5] and Optimal Partitioning [6] formulate the problem as the minimization of a sum of segment-wise losses plus a penalty on the number of changepoints. By exploiting the optimal substructure of the problem, these methods guarantee global optimality and admit polynomial-time algorithms under suitable assumptions.

Substantial effort has been devoted to improving the practical efficiency of DP-based CPD. Notable examples include the Pruned Exact Linear Time (PELT) algorithm [7], which uses inequality-based pruning to reduce the search space, and functional pruning approaches [8,9], which exploit convexity properties of the segment cost. These methods have become state of the art for large-scale problems with additive objectives.

Despite their efficiency, DP-based models fundamentally rely on the separability of the objective function across segments. This reliance severely limits their ability to incorporate constraints that couple decisions across the entire segmentation. While constrained DP and state-augmented formulations have been proposed for specific problems [10], the resulting state spaces typically grow exponentially, making such approaches impractical beyond very limited forms of global control. Consequently, DP methods remain best suited for problems with simple additive objectives and limited structural constraints.

Several optimal CPD models reinterpret the segmentation problem as a shortest-path problem on a directed acyclic graph, where nodes correspond to time indices and arcs represent feasible segments weighted by their associated costs [11,12]. This representation is mathematically equivalent to DP and enables the use of efficient shortest-path algorithms.

While graph-based formulations provide useful conceptual insight and algorithmic flexibility, they inherit the same structural limitations as DP. In particular, global constraints that do not decompose additively along a path are difficult to express without introducing additional dimensions or complex side constraints, which again compromise tractability.

Mixed-Integer Programming (MIP) has been previously applied to changepoint detection and piecewise regression, primarily through formulations based on breakpoint indicator variables or segment assignment variables. Early work on segmented regression using integer programming can be traced back to [13] and [14], with more recent formulations incorporating modern MIP solvers for exact piecewise linear fitting [15].

In breakpoint-based formulations, binary variables indicate whether a changepoint occurs at a given time index, while continuous variables model segment-specific parameters. These models allow regression and segmentation decisions to be optimized jointly and can accommodate certain side constraints. However, constraints involving segment-level properties—such as length distributions, aggregate statistics of selected segments, or interactions between non-adjacent segments—typically require additional auxiliary variables and complex logical constraints, leading to weak linear relaxations and limited scalability.

Assignment-based formulations, in which observations are assigned to segments via binary variables, face similar difficulties when enforcing contiguity and global structural constraints. In both cases, segmentation is encoded implicitly, rather than treating segments themselves as first-class decision objects.

The approach proposed in this paper differs fundamentally from the above paradigms by adopting a set-based perspective. Each feasible segment is explicitly represented as a decision variable, and the segmentation problem is formulated as a set covering problem in which the time axis must be covered exactly once. While set covering and set partitioning models are classical tools in combinatorial optimization [16,17], they have received comparatively little attention in the changepoint detection literature.

By elevating segments to decision variables, the set-covering formulation provides a natural mechanism for expressing global structural constraints directly at the segment level. Aggregate constraints on the number, length, distribution, or interaction of segments can be imposed through simple linear inequalities, without resorting to state augmentation or intricate logical constructs. This distinguishes the proposed SC-CPD framework from existing MIP formulations and enables a level of modeling expressiveness that is difficult to achieve within DP-based, shortest-path, or breakpoint-indicator models.

In summary, existing optimal CPD models exhibit a clear trade-off between computational efficiency and modeling flexibility. Dynamic programming and graph-based approaches provide excellent performance for additive objectives but offer limited support for global structural constraints. Prior MIP formulations increase modeling power but typically encode segmentation implicitly and encounter scalability issues when complex segment-level constraints are introduced.

The proposed SC-CPD framework occupies a complementary position in this landscape. By combining a set-covering formulation with modern MILP techniques and problem-specific preprocessing, it delivers global optimality while substantially expanding the class of constraints that can be handled in a principled and scalable manner. This positions SC-CPD as a flexible and extensible optimization framework for changepoint detection in applications where global structure and domain-specific requirements play a central role.

For a MIQP quadratic formulation see [18]

2.1. Global series constraints

The literature proposed many constraints that could be requested on the single segments of the model or on the whole modelled series. Typical constraints on single segments include

- Minimum spacing between changepoints: Enforce a global minimum gap (e.g., at least L points between any two changepoints) to prevent over-fragmentation of the series.
- Boundary exclusion: Disallow changepoints too close to the start or end (e.g., first/last K points), ensuring stable estimation at boundaries.
- Seasonal/calendar alignment: Restrict changepoints to a predefined set of admissible positions (e.g., month-ends, business-cycle markers), imposing global positional structure.
- Equal or bounded segment sizes: Require near-equal segment lengths or bound length variability (e.g., max/min segment length ratio), affecting the global partition shape.
- Group sparsity on positions: Impose structured sparsity over candidate positions (e.g., block or clustered positions), selecting changepoints from predefined groups globally.
- Template matching: Require changepoint positions to follow a given motif (e.g., known intervention schedule) with limited deviations across the whole timeline.

Many changepoint frameworks encode these ideas via global penalization rather than hard constraints—using penalties on the number of changepoints (*L0*), total variation (*L1*), or graph-structured transitions that enforce sign alternation or monotone regimes across the entire sequence. These approaches provide a way to control global behaviors, such as roughness or regime structure, and are usually implemented by dynamic programming or specialized algorithm limiting the possibility to flexibly expand the proposed model.

These are the constraints that "operate over the whole segmentation" and are discussed in the global constraint catalogues and CP-based changepoint detection literature (Beldiceanu, Régin; O'Hearn; Gionis; Killick; etc.), as well as in bioinformatics segmentation and shape-constrained segmentation. The main global constraints identified in the segmentation literature (CP + changepoint detection + structural break models) are:

- Segment-count constraints (you already use);
- Constraints on the distribution of segment lengths (global AMONG/NVALUES)
- Constraints on the global pattern of changes (REGULAR, PEAK/VALLY count)
- Global monotonicity or global shape constraints
- Global cost budgets / global total variation constraints
- Global constraint on number of distinct regimes
- Global maxima/minima on segment attributes (e.g., amplitude range)
- Global forbidden-pattern constraints
- Global window-based count constraints
- Global constraints on total curvature or total slope

Constraints for changepoint sets (MERGE WITH ABOVE)

- Minimum/maximum number of changepoints: ensure at least / at most a certain number to avoid under-segmentation in known nonstationary contexts.
- Total variation budget: Constrain the sum of absolute jump magnitudes across all changepoints to be below a threshold, controlling global roughness.
- Net drift constraint: Enforce that the signed sum of jumps equals a specified value (e.g., zero), ensuring the process starts/ends at comparable levels. Endpoint level constraints: Constrain the final level to lie within a target range relative to the initial level (e.g., bounded net change), coupling start/end globally.
- Maximum total jump magnitude: Cap the aggregate absolute change across all changepoints, distinct from the count of changepoints.
- Monotonic episode limits: Allow only a bounded number of monotone episodes (e.g., at most M runs of nondecreasing levels), acting globally over segments.
- Parity or count constraints: Enforce an even/odd number of changepoints, or fixed counts of specific jump types (e.g., exactly R upward jumps).

- Total penalty budget: Instead of a fixed penalty per changepoint, constrain the sum of penalties (data-adaptive or position-weights), globally bounding the objective's regularization.
- Symmetry constraints: Enforce symmetry of changepoint locations around the series midpoint, or mirrored jump magnitudes, for symmetric processes.
- Multi-scale consistency: Enforce that changepoints persist across coarse-to-fine aggregations (e.g., daily to weekly), selecting positions robust globally.
- Robustness-to-outliers budget: Limit the number of changepoints attributable to isolated spikes by bounding the cumulative outlier-attributed evidence globally.

3. An extended set covering model

The core idea of the model presented here is to list all subsequences of series values that satisfy specific structural constraints, in the following denoted as *feasible runs* of values, quantify a quality measure of each run, and select the subset of runs that collectively cover the entire series while minimizing a cost of the difference between actual and model data (the model *residuals*) or maximizing a model fitness measure. This is the core of the model, further constraints can be added to accommodate specific desiderata on the model, taking advantage of the modeling flexibility of mixed-integer programming and the effectiveness of state-of-the-art general MIP solvers [19].

A Set Partitioning Problem (SPP) is at the core of the model, where the objective is to minimize the cost of the residuals and the constraints ensure that each point of the series has a corresponding one in the model. Note that no assumption is made about the nature of the model of the segments, it can be linear regression, giving rise to piecewise linear regression, but also any other nonlinear model. It is also possible to use different models for different runs at no cost to the optimization process. The SPP is defined over a set $X = [x_j]$, $j = 1, \ldots, nruns$ of binary decision variables, each associated with one of the runs, i.e., of the segments. The generation of feasible runs can already implement some dominance, e.g. avoiding the generation of runs with too few or too many values. The cost c_j of each run can be computed according to any of the fitness measures proposed in the literature, including R^2, SER, χ^2, RMSE, simple variance etc. The constraints ensure that each point of the series is covered by a selected run.

Unfortunately, the number of feasible runs for a real-world application can be very large, and the SPP can take an unacceptable amount of time to solve. Therefore, we relax the equality constraints into inequalities, transforming the problem into a Set Covering Problem (SCP), which can typically be solved in much higher dimensions, at the cost of having multiple runs covering some of the series points, therefore requiring a postprocessing to get a feasible segmentation. The resulting TSSC (standing for Time Series Set Covering) model is the following, where coefficients a_{ij} are 0/1 coefficients taking the value of 1 if and only if run j covers the i-th series point, $i \in \{1, \ldots, npoints\}$ and $j \in \{1, \ldots nruns\}$.

$$(TSSC) \qquad z_{TSSC} = \min \sum_{j=1}^{nruns} c_j x_j \tag{1}$$

$$s.t. \sum_{j=1}^{nruns} a_{ij} x_j \geq 1, \qquad i = 1, \ldots, npoints \tag{2}$$

$$\sum_{j=1}^{nruns} x_j \leq maxruns \tag{3}$$

$$x_j \in \{0, 1\}, \qquad j = 1, \ldots, nruns \tag{4}$$

State-of-the-art MIP solvers are very effective on SCP, but very large instances can still require high computational times. We have therefore also implemented a Lagrangian matheuristic [20] on the TSSC problem, relaxing all constraints (2) by associating a Lagrangian penalty λ_i to each i-th constraint, $1 \leq i \leq npoints$, and keeping only constraint (3) in the model. This results in model LTSSC (standing for Lagrangian TSSC) and we solved it by a subgradient algorithm, where the subproblem is very easy to solve, requiring to select at most the $maxruns$ most negative variables at each subgradient iteration. A simple fixing heuristic takes the incumbent subproblem solution at each iteration, which can be infeasible because it does not cover all relaxed constraints, and adds selected variables until it becomes feasible.

4. Computational experience

We implemented models TSSC and LTSSC in C++ and ran them on a Windows 11 Intel i7 machine equipped with 32 Gb of RAM. The MIP solver used was CPLEX 22.11. All codes and data are available from the project repository.

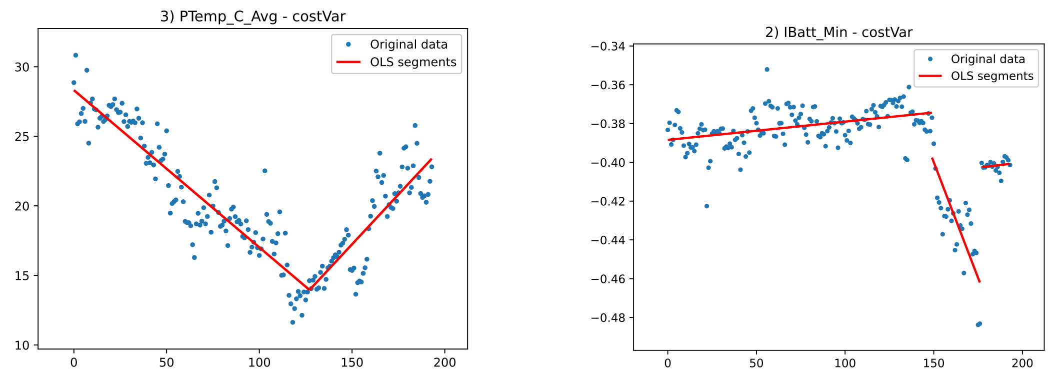
The computational results reported here, still to be considered as preliminary, are mainly obtained on environmental monitoring data series, which initially prompted this research. The series were produced in the context of the SMARTLAGOON project, an EU H2020 project, born with the primary objective of developing a tool to allow real-time monitoring, analysis, to predict socio-environmental evolution of the vulnerable area *Mar Menor*, which is the largest saltwater coastal lagoon in Europe. [21]. Within the framework of the project, smart buoys were deployed in the lagoon and their sensors generated series on 12 attributes.

The sensed variables used here are: the steam pressure (Vapor-Pressure-Avg), the average wind speed (WS-ms-Avg), water temperature measured by a thermistor at the depth of 0.5m (ThermTemp1-Avg), water temperature measured by oximeter at different depths (1m - Wtemp-C1-Avg, 3m - Wtemp-C2-Avg), water temperature measured by conductimeter at the depth of 1m (SDI-Temp-1m), minimum current in the battery (IBatt-Min), average and maximum temperature obtained by a datalogger panel temp thermocouple (PTemp-C-Avg, PTemp-C-Max) and average charging voltage (V-in-chg-Avg). The overall dataset was recorded from August 2022 to April 2023. We complemented these datasets with two others from different domains: economics (bitcoin-US dollar exchange rate, BTC-USD) and healthcare (COVID infections in Italy, Covid-Italia-22) to get a first indication of the generality of the approach across domains.

Table 1. Data series results

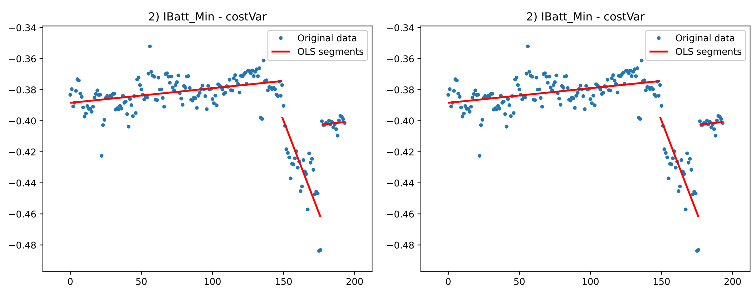
Dataseries	*npoints*	*nruns*	*nsegm*	*truns*	*tsolve*	*QMSE*	*QMSE1*	*QMSE2*
Vapor-Pressure-Avg	194	15400	6	0	1	0.90	1.14	1.03
Vapor-Pressure-Avg-2	1139	618828	7	30	832	2.85	3.26	3.32
WS-ms-Avg	1139	618828	2	30	1136	162	193.03	210.80
ThermTemp1-Avg	194	15400	8	0	1	5.12	5.98	na*
Wtemp-C1-Avg	1139	618828	19	30	1212	13.89	22.63	na*
Wtemp-C2-Avg	194	15400	8	0	1	50.75	68.15	na*
SDI-Temp-1m	1139	618828	21	30	1354	15.06	25.06	na*
IBatt-Min	194	15400	3	0	1	3.19e-3	5.29e-3	3.27e-3
PTemp-C-Avg	194	15400	5	0	1	46.40	68.56	58.12
PTemp-C-Max	194	15400	4	0	1	114.24	139.61	125.09
V-in-chg-Avg	194	15400	2	0	1	9.20	9.38	9.38
BTC-USD	366	60378	12	3	4	1.88e7	7.51e7	2.42e7
Covid-Italia-22	507	109746	5	-	11	5070.95	na	na

The results are summarized in table 1. The columns show the name of the series (*Dataseries*), the number of data points (*npoints*), the number of generated runs (*nruns*), the number of chosen segments (*nsegm*), the CPU time in seconds to generate all runs (*truns*), the CPU time in seconds to generate all runs (*truns*), the CPU time to solve the extended SCP model (*tsolve*), and a *quasi RMSE* quality measure of the solutions obtained by the algorithm described in section 3, *QMSE*, and, for comparison, the same measure for the solutions obtained by the codes of [22], *QMSE1*, and of [4], *QMSE2*. A comparison with the results from *Ruptures* [3] is being finalized and will be presented at the conference. Data with asterisks require special comments, which are incompatible with the page limit of this note, but will be presented at the conference and in the full version of the paper, *na*'s mean that no feasible solution was produced.

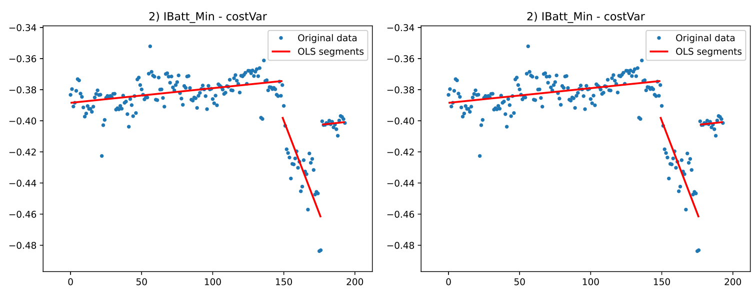


(**a**) Average temperature (**b**) Minimum battery current

Figure 1. Environmental dataseries



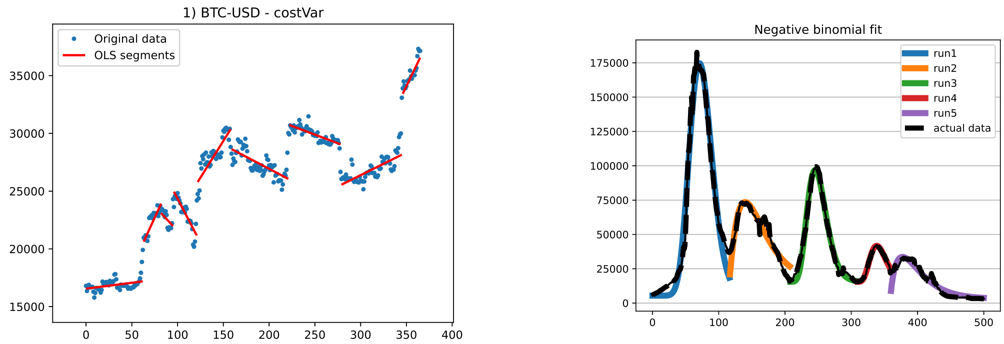
(**a**) (**b**)



(**c**) (**d**)

Figure 2. This is a wide figure. Schemes follow the same formatting. If there are multiple panels, they should be listed as: (**a**) Description of what is contained in the first panel. (**b**) Description of what is contained in the second panel. (**c**) Description of what is contained in the third panel. (**d**) Description of what is contained in the fourth panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.

Not surprisingly, the set partitioning model could always produce the better solution, as it was the only algorithm that explicitly used the proposed cost function for optimization. More interesting is the limited computational time needed to solve instances with up to a few hundred variables, while the "curse of dimensionality" becomes apparent when solving instances with more than 1000 variables. The healthcare instance, here only a validation case, proves that the approach is also effective for nonlinear segmentation, but the search for optimal fitting parameters requires a very high CPU time. The figures 1 show the solution for two environmental data series, while the figures 3 show the two non environmental data series.



(**a**) Bitcoin - USD exchange rate

(**b**) Covid Italy 2022

Figure 3. Economics and healthcare dataseries

5. Conclusions

The paper presents an MIP-based approach, both a Lagrangian matheuristic and an exact model, to time series segmentation. The underlying mathematical model is able to adapt to different requirements on the resulting model, making it more tolerant to residuals or more representative of short trends. The analysis imposes no constraints on the model associated with each segment, which can be linear, polynomial, or defined on any distribution function of interest. It can even be a combination of different models, allowing for linearity on some segments and, for example, exponential increases on others.

This flexibility comes at the cost of having to solve a NP problem on large size instances. The increase in effectiveness of general MIP solvers allows to consider the solution of real-world instances to optimality, and it also provides the basis for the design of mathematically grounded heuristics. We report preliminary results on sensor data series obtained in an environmental monitoring, but also two experiments on financial and healthcare use cases.

The results so far provide only a first indication of the possibilities offered by mathematical models, but also of the difficulties to be overcome. Besides the obvious limit imposed by the NP-hardness of the problem (the "curse of dimensionality"), which is only partially alleviated by heuristic solving, there are problem-specific issues to be faced, such as which cost function to use or which constraint to impose on the runs. In fact, the final result is only indirectly determined by our model, and there are still cases where results are unsatisfactory to the eye, even though they are numerically satisfying. Current results have already proven useful in the motivating application domain, both for missing value imputation and for short-term forecasting, obtained by decomposing the series down to the last segment and using the identified components to extend it into the near future. Moreover, these results do not seem to be affected by the application domain.

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Acknowledgments: "During the preparation of this manuscript/study, the author(s) used GenAI only for the purposes of syntax and style checking."

Conflicts of Interest: "The authors declare no conflicts of interest."

Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

Table A1. This is a table caption.

Title 1	**Title 2**	**Title 3**
Entry 1	Data	Data
Entry 2	Data	Data

Appendix B

All appendix sections must be cited in the main text. In the appendices, Figures, Tables, etc. should be labeled, starting with "A"—e.g., Figure A1, Figure A2, etc.

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